

Who's to Blame for Survey Instability: Respondents with Nonexistent Preferences or Researchers with Flawed Measures?¹

Gary King²

Institute for Quantitative Social Science
Harvard University

Society for Political Methodology, 7/18/2026

¹Slides, paper, data: [GaryKing.org/instability](https://garyking.org/instability)

²Joint work with [Libby Jenke](#), University of Houston

A Quiz (for you, as a survey researcher)

A Quiz (for you, as a survey researcher)

- Ask: any nontrivial, binary choice survey question

A Quiz (for you, as a survey researcher)

- **Ask:** any nontrivial, binary choice survey question
- **Ask again:** same question

A Quiz (for you, as a survey researcher)

- Ask: any nontrivial, binary choice survey question
- Ask again: same question
- What % answer differently?

A Quiz (for you, as a survey researcher)

- Ask: any nontrivial, binary choice survey question
 - Assume: No (1) memory, (2) material changes, (3) attrition
- Ask again: same question
- What % answer differently?

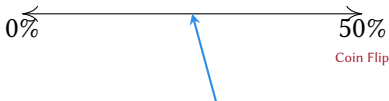
A Quiz (for you, as a survey researcher)

- **Ask:** any nontrivial, binary choice survey question
 - **Assume:** No (1) memory, (2) material changes, (3) attrition
- **Ask again:** same question

- **What % answer differently?** 0%  50%
Coin Flip

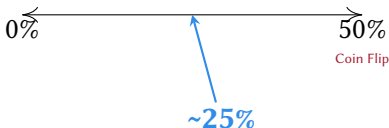
A Quiz (for you, as a survey researcher)

- **Ask:** any nontrivial, binary choice survey question
 - **Assume:** No (1) memory, (2) material changes, (3) attrition
- **Ask again:** same question
- **What % answer differently?**



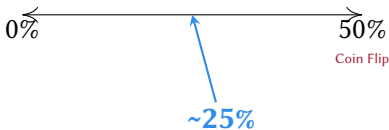
A Quiz (for you, as a survey researcher)

- **Ask:** any nontrivial, binary choice survey question
 - **Assume:** No (1) memory, (2) material changes, (3) attrition
- **Ask again:** same question
- **What % answer differently?**



A Quiz (for you, as a survey researcher)

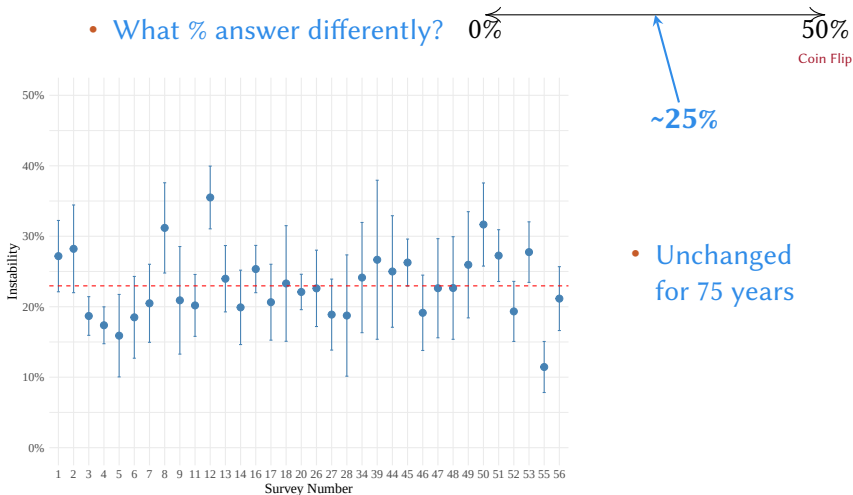
- **Ask:** any nontrivial, binary choice survey question
 - **Assume:** No (1) memory, (2) material changes, (3) attrition
- **Ask again:** same question
- **What % answer differently?**



- **Unchanged**
for 75 years

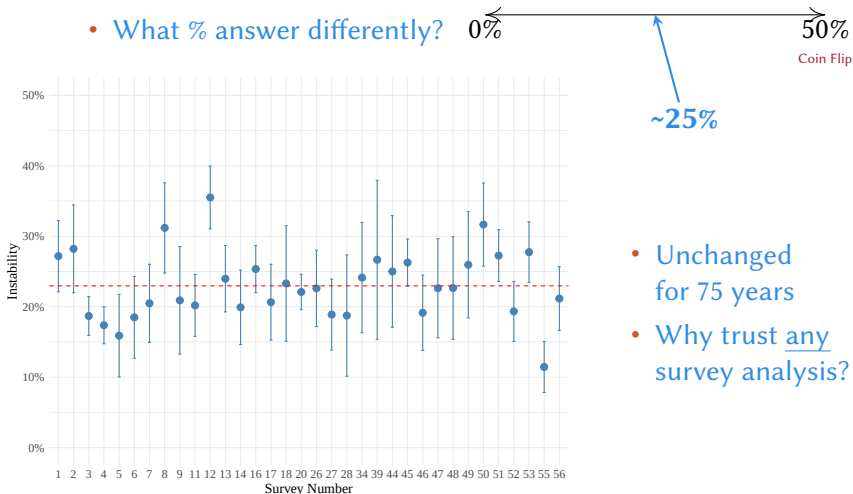
A Quiz (for you, as a survey researcher)

- **Ask:** any nontrivial, binary choice survey question
 - **Assume:** No (1) memory, (2) material changes, (3) attrition
- **Ask again:** same question
- **What % answer differently?**



A Quiz (for you, as a survey researcher)

- **Ask:** any nontrivial, binary choice survey question
 - **Assume:** No (1) memory, (2) material changes, (3) attrition
- **Ask again:** same question
- **What % answer differently?**



- Unchanged for 75 years
- Why trust any survey analysis?

Research Strategy

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
- New Experimental Platform

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
- New Experimental Platform

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2 

A Binary Choice Conjoint Question

(Ono and Burden, 2018)

Please carefully review the two potential candidates running for election to the U.S. House of Representatives, detailed below.

	Candidate 0	Candidate 1
Race/Ethnicity	Hispanic	Asian American
Age	52	60
Favorability rating among the public	70%	34%
Position on immigrants	Favors giving citizenship or guest worker status to undocumented immigrants	Opposes giving citizenship or guest worker status to undocumented immigrants
Party affiliation	Republican Party	Democratic Party
Position on abortion	Abortion is not a private matter (pro-life)	Abortion is a private matter (pro-choice)
Position on government deficit	Wants to reduce the deficit through tax increase	Wants to reduce the deficit through tax increase
Salient personal characteristics	Really cares about people like you	Really cares about people like you
Position on national security	Wants to cut military budget and keep the U.S. out of war	Wants to maintain strong defense and increase U.S. influence
Gender	Female	Female
Policy area of expertise	Education	Foreign policy
Family	Single (divorced)	Married (no child)
Experience in public office	12 years	4 years

If you had to choose between them, which of these candidates would you vote to be a member of the U.S. House of Representatives?

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2 

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2
 - Assumptions: Easy: (2) and (3). Also works: (1).

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2
 - Assumptions: Easy: (2) and (3). Also works: (1). 

A Binary Choice Conjoint Question

(Ono and Burden, 2018)

Please carefully review the two potential candidates running for election to the U.S. House of Representatives, detailed below.

	Candidate 0	Candidate 1
Race/Ethnicity	Hispanic	Asian American
Age	52	60
Favorability rating among the public	70%	34%
Position on immigrants	Favors giving citizenship or guest worker status to undocumented immigrants	Opposes giving citizenship or guest worker status to undocumented immigrants
Party affiliation	Republican Party	Democratic Party
Position on abortion	Abortion is not a private matter (pro-life)	Abortion is a private matter (pro-choice)
Position on government deficit	Wants to reduce the deficit through tax increase	Wants to reduce the deficit through tax increase
Salient personal characteristics	Really cares about people like you	Really cares about people like you
Position on national security	Wants to cut military budget and keep the U.S. out of war	Wants to maintain strong defense and increase U.S. influence
Gender	Female	Female
Policy area of expertise	Education	Foreign policy
Family	Single (divorced)	Married (no child)
Experience in public office	12 years	4 years

If you had to choose between them, which of these candidates would you vote to be a member of the U.S. House of Representatives?

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2
 - Assumptions: Easy: (2) and (3). Also works: (1). 

Research Strategy

- Assumptions: No (1) memory, (2) material changes, (3) attrition
- Previous Research
 - Strategy: separate surveys, months apart
 - Problems: the 3 assumptions conflict
- New Experimental Platform
 - Single survey, conjoint question: Q_1 [distractors] Q_2
 - Assumptions: Easy: (2) and (3). Also works: (1).
 - New Technology: online survey platforms

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)
 - **We were wrong & changed directions:**

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)
 - **We were wrong & changed directions:** A lot.

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)
 - **We were wrong & changed directions:** A lot.
 - **The 3 R's of Transparency**

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)
 - **We were wrong & changed directions:** A lot.
 - **The 3 R's of Transparency**
 - **Replicability:** data & code in Dataverse

Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)
 - **We were wrong & changed directions:** A lot.
 - **The 3 R's of Transparency**
 - **Replicability:** data & code in Dataverse
 - **Reliability:** easy to collect new data from scratch

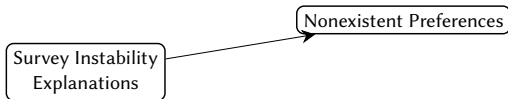
Research Strategy

- **Assumptions:** No (1) memory, (2) material changes, (3) attrition
- **Previous Research**
 - **Strategy:** separate surveys, months apart
 - **Problems:** the 3 assumptions conflict
- **New Experimental Platform**
 - **Single survey, conjoint question:** Q_1 [distractors] Q_2
 - **Assumptions:** **Easy:** (2) and (3). **Also works:** (1).
 - **New Technology:** online survey platforms
 - **Prep time:** months $\rightsquigarrow$ hours
 - **Validation:** Preregistration + one test $\rightsquigarrow$ many sequential tests
 - **Stopping rule:** 1 run $\rightsquigarrow$ ability to predict in new data
 - **We tested:** every prior hypothesis & ours, in many ways
 - **We ran:** 59 unique surveys (observational, experimental, eye tracking, open/closed ended, conjoint/standard,)
 - **We were wrong & changed directions:** A lot.
 - **The 3 R's of Transparency**
 - **Replicability:** data & code in Dataverse
 - **Reliability:** easy to collect new data from scratch
 - **Reporting:** every failed hypothesis & mistake

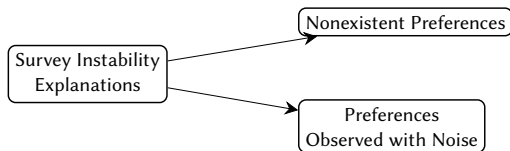
What We Know

Survey Instability
Explanations

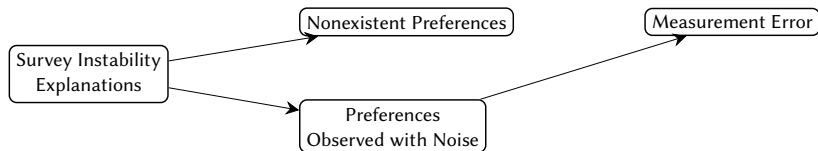
What We Know



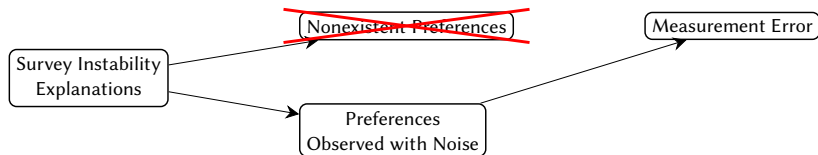
What We Know



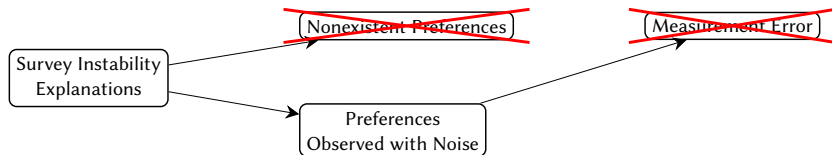
What We Know



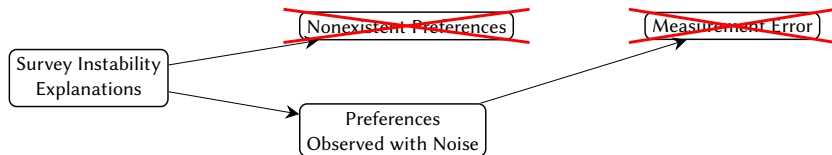
What We Know



What We Know

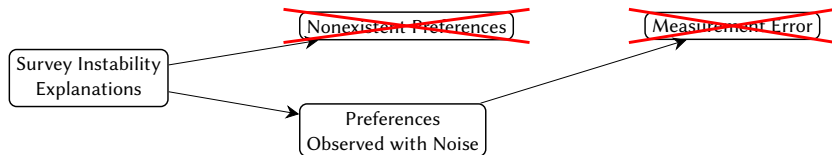


What We Know



- Nonexistent Preferences

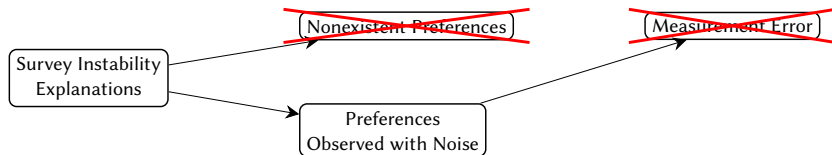
What We Know



- Nonexistent Preferences

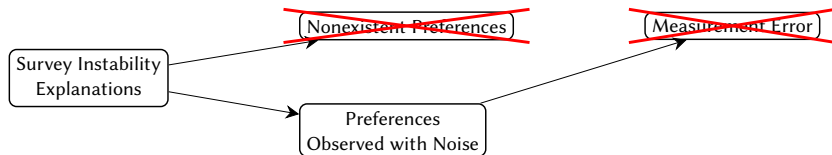
- Measurement Error

What We Know



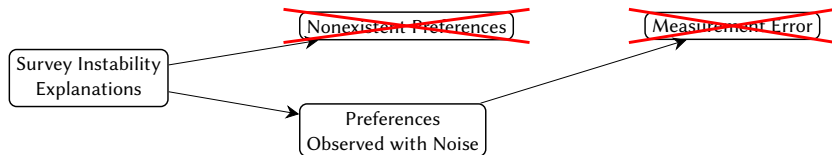
- Nonexistent Preferences
 - ~all evidence contradicts Converse
- Measurement Error

What We Know



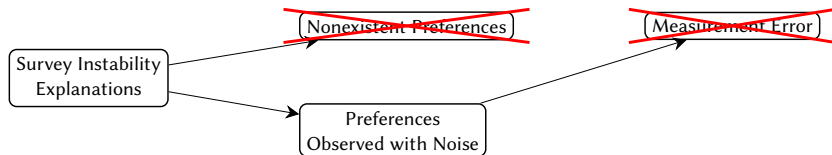
- **Nonexistent Preferences**
 - ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
- **Measurement Error**

What We Know



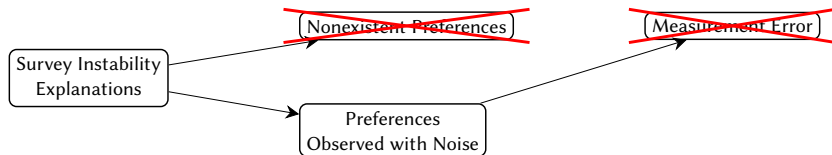
- **Nonexistent Preferences**
 - ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
 - **Persuasion attempts:** usually fail
- **Measurement Error**

What We Know



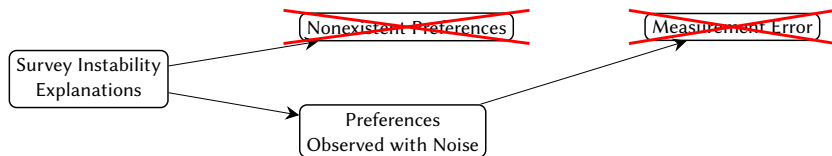
- **Nonexistent Preferences**
 - ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
 - **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- **Measurement Error**

What We Know



- **Nonexistent Preferences**
 - ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
 - **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
 - **⇒ Preferences exist & rarely change (without a reason)**
- **Measurement Error**

What We Know

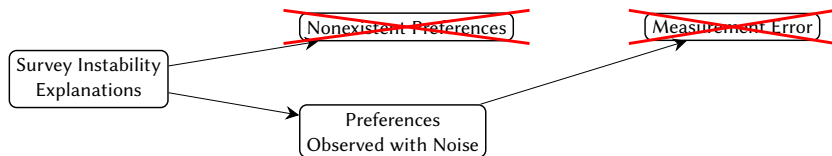


- ~~Nonexistent Preferences~~

- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- $\Rightarrow$ Preferences exist & rarely change (without a reason)

- Measurement Error

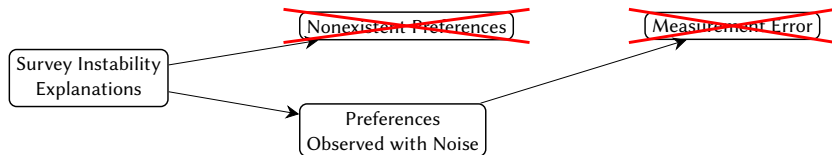
What We Know



~~Nonexistent Preferences~~

- ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
 - **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
 - $\Rightarrow$ Preferences exist & rarely change (without a reason)
- ## Measurement Error
- **The Literature:** must be M.E., because people have preferences!

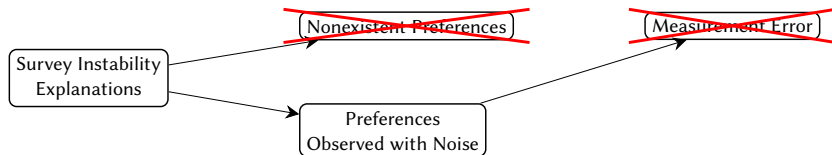
What We Know



~~Nonexistent Preferences~~

- ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
 - **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
 - $\Rightarrow$ Preferences exist & rarely change (without a reason)
- ## Measurement Error
- **The Literature:** must be M.E., because people have preferences!
 - **Survey improvements:**

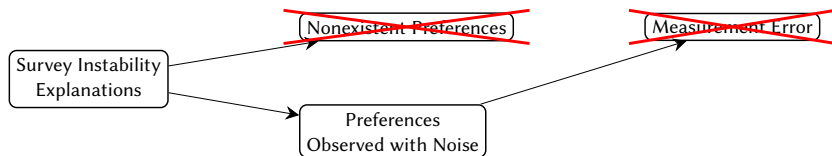
What We Know



~~Nonexistent Preferences~~

- ~all evidence contradicts Converse
 - Survey findings: numerous, strong, replicable
 - **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
 - $\Rightarrow$ Preferences exist & rarely change (without a reason)
- ## Measurement Error
- **The Literature:** must be M.E., because people have preferences!
 - **Survey improvements:**
 - affect both questions

What We Know



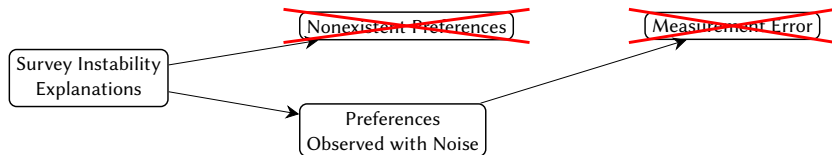
~~• Nonexistent Preferences~~

- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- $\Rightarrow$ Preferences exist & rarely change (without a reason)

• Measurement Error

- **The Literature:** must be M.E., because people have preferences!
- **Survey improvements:**
 - affect both questions
 - reduce bias, not instability

What We Know



~~• Nonexistent Preferences~~

- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- $\Rightarrow$ Preferences exist & rarely change (without a reason)

• Measurement Error

- **The Literature:** must be M.E., because people have preferences!
- **Survey improvements:**
 - affect both questions
 - reduce bias, not instability 

What We Know

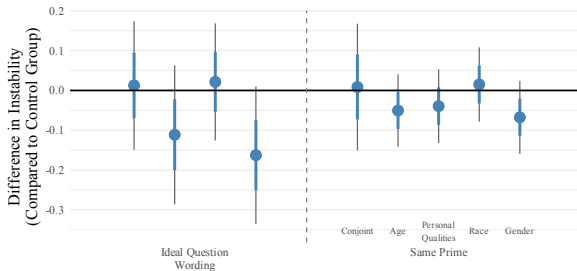
~~Nonexistent Preferences~~

~~Measurement Error~~

Survey Instability

Explains

Reducing Bias Doesn't Reduce Instability



- reduce bias, not instability



What We Know

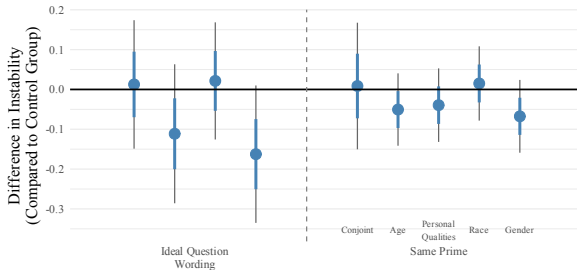
~~Nonexistent Preferences~~

~~Measurement Error~~

Survey Instability

Explains

Reducing Bias Doesn't Reduce Instability



- Left: 2008 ANES question wording changes

- reduce bias, not instability



What We Know

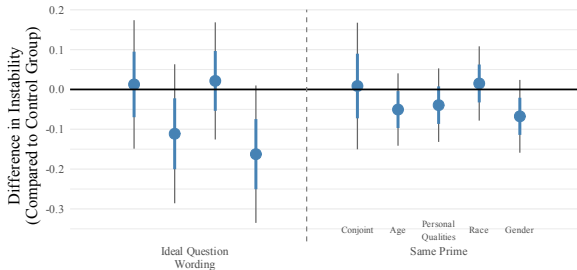
~~Nonexistent Preferences~~

~~Measurement Error~~

Survey Instability

Explains

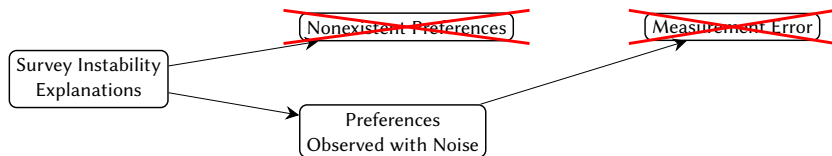
Reducing Bias Doesn't Reduce Instability



- Left: 2008 ANES question wording changes
- Right: different (priming) “considerations”
- reduce bias, not instability



What We Know



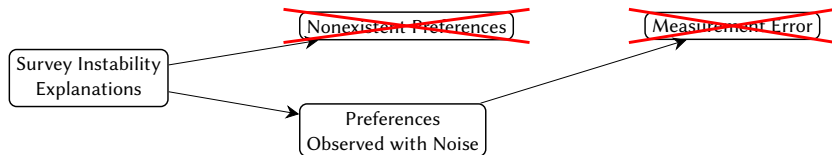
~~• Nonexistent Preferences~~

- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- **⇒ Preferences exist & rarely change (without a reason)**

• Measurement Error

- **The Literature:** must be M.E., because people have preferences!
- **Survey improvements:**
 - affect both questions
 - reduce bias, not instability 

What We Know



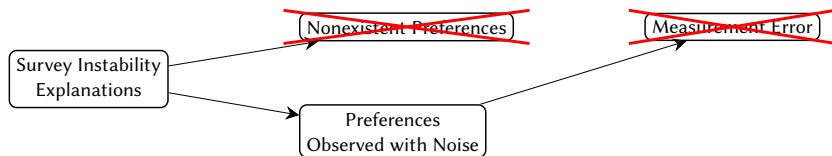
~~Nonexistent Preferences~~

- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- $\Rightarrow$ Preferences exist & rarely change (without a reason)

Measurement Error

- **The Literature:** must be M.E., because people have preferences!
- **Survey improvements:**
 - affect both questions
 - reduce bias, not instability
- $\Rightarrow$ Most instability isn't due to measurement error

What We Know



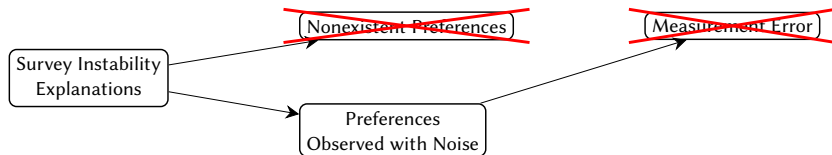
~~Nonexistent Preferences~~

- ~all evidence contradicts Converse
- Survey findings: numerous, strong, replicable
- **Persuasion attempts:** usually fail
 - **Candidates, dictators, marketers:** change subject, not preferences
- $\Rightarrow$ Preferences exist & rarely change (without a reason)

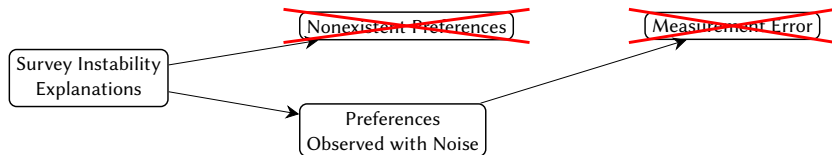
~~Measurement Error~~

- **The Literature:** must be M.E., because people have preferences!
- **Survey improvements:**
 - affect both questions
 - reduce bias, not instability
- $\Rightarrow$ **Most instability isn't due to measurement error**

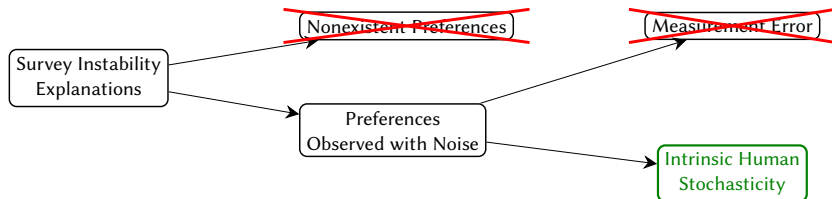
What We Know



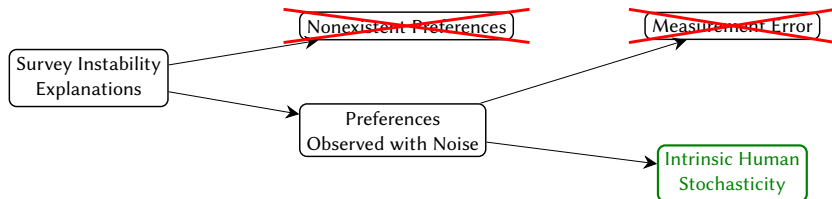
What We Show



What We Show

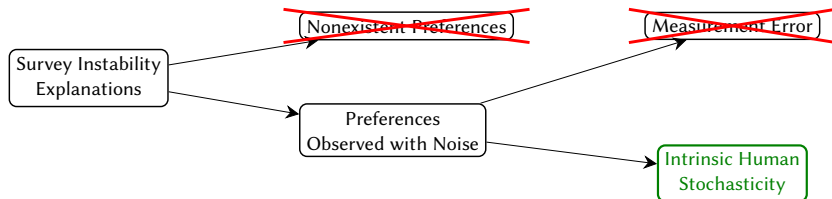


What We Show



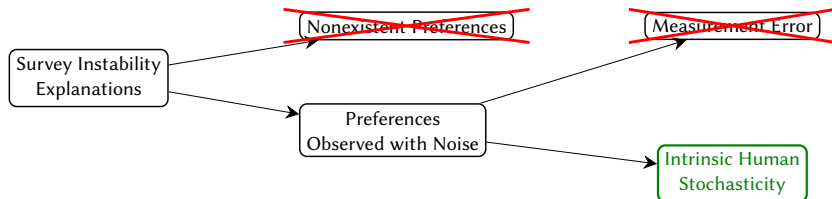
- Generally, in other fields: instrument v. subject variability

What We Show



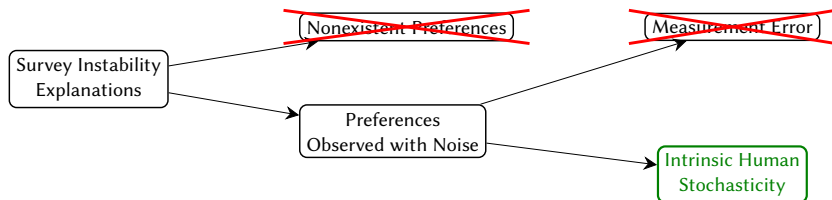
- Generally, in other fields: instrument v. subject variability
 - soft matter physics: tracking error v. thermal noise

What We Show



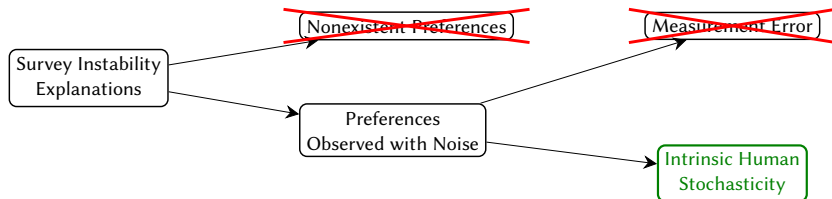
- Generally, in other fields: instrument v. subject variability
 - psychometrics: measurement error v. individual differences

What We Show



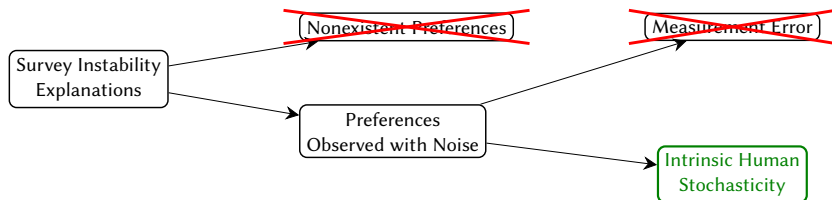
- Generally, in other fields: instrument v. subject variability
 - biology, medicine: technical v. biological variability

What We Show



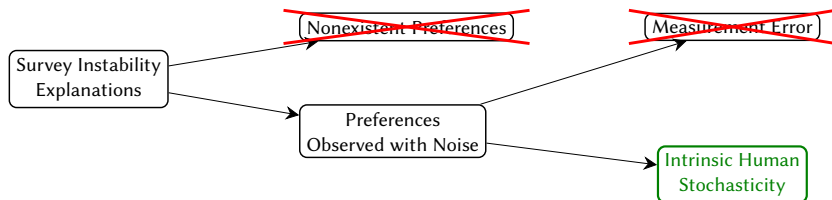
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty

What We Show



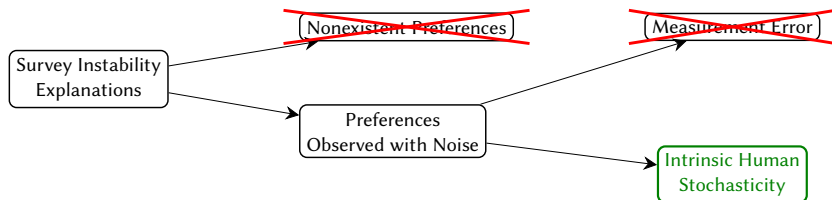
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions

What We Show



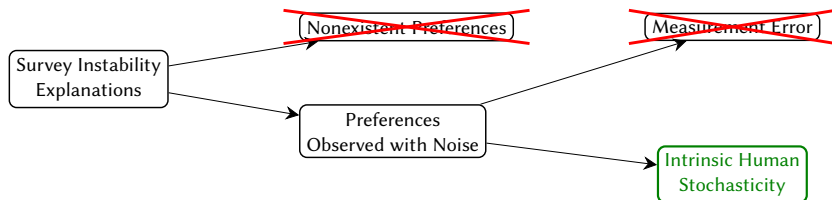
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions
 - “**Intrinsic**”: A component of; essential to its existence

What We Show



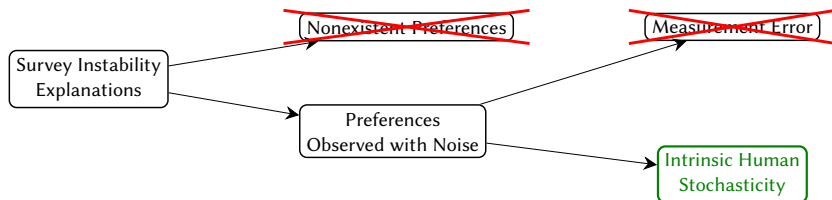
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions
 - “**Intrinsic**”: A component of, essential to its existence
 - **Zero stochasticity**: not an attentive respondent, or live human

What We Show



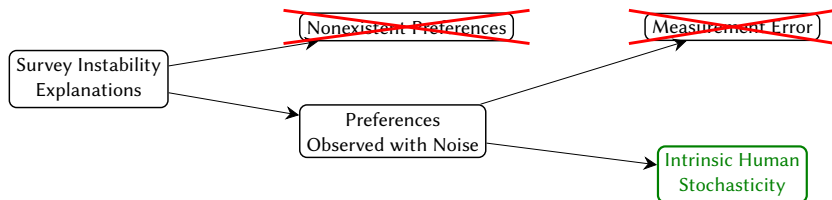
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions
 - “**Intrinsic**”: A component of, essential to its existence
 - **Zero stochasticity**: not an attentive respondent, or live human
- Imagine 2 CT Scans moments apart

What We Show



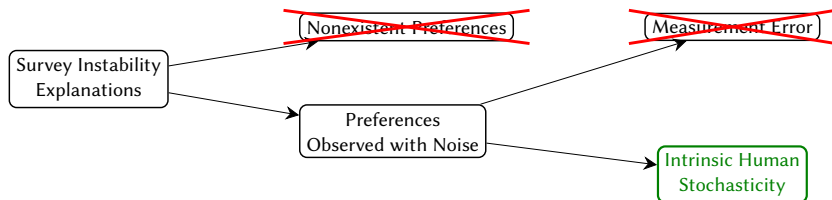
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions
 - “**Intrinsic**”: A component of; essential to its existence
 - **Zero stochasticity**: not an attentive respondent, or live human
- Imagine 2 CT Scans moments apart
 - **Everything changes**: Circulation, digestion, signaling, heart, stomach, hormones, synapses, eyes; 1.2% cells/day replaced, etc.

What We Show



- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions
 - “**Intrinsic**”: A component of; essential to its existence
 - **Zero stochasticity**: not an attentive respondent, or live human
- Imagine 2 CT Scans moments apart
 - **Everything changes**: Circulation, digestion, signaling, heart, stomach, hormones, synapses, eyes; 1.2% cells/day replaced, etc.
 - **Stochasticity’s a feature**: improves learning, creativity, stability

What We Show



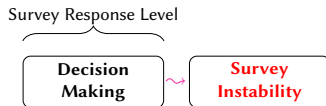
- Generally, in other fields: **instrument** v. **subject** variability
 - social science methods: **estimation** v. **fundamental** uncertainty
- Definitions
 - “**Intrinsic**”: A component of; essential to its existence
 - **Zero stochasticity**: not an attentive respondent, or live human
- **Imagine 2 CT Scans moments apart**
 - **Everything changes**: Circulation, digestion, signaling, heart, stomach, hormones, synapses, eyes; 1.2% cells/day replaced, etc.
 - **Stochasticity’s a feature**: improves learning, creativity, stability

⇒ **The creatures we study**: variables, not constants

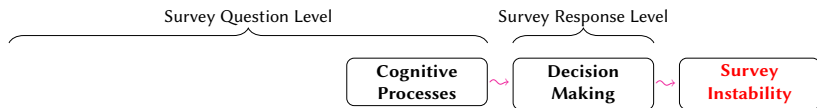
Sources of Stochasticity Leading to Survey Instability

**Survey
Instability**

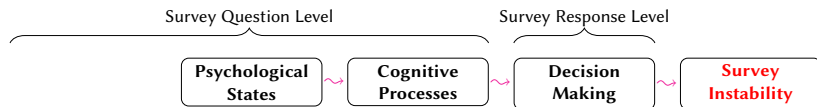
Sources of Stochasticity Leading to Survey Instability



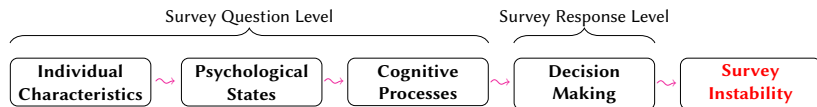
Sources of Stochasticity Leading to Survey Instability



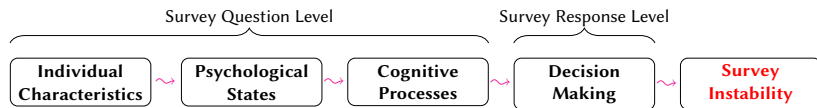
Sources of Stochasticity Leading to Survey Instability



Sources of Stochasticity Leading to Survey Instability

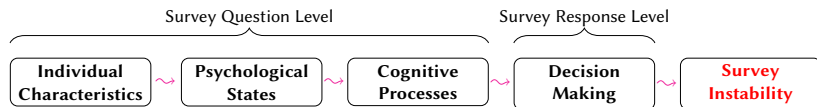


Sources of Stochasticity Leading to Survey Instability



Used to find

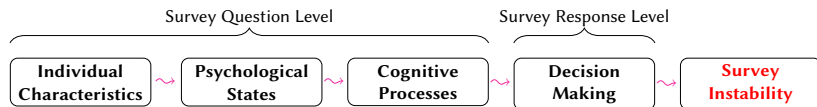
Sources of Stochasticity Leading to Survey Instability



Used to find

- **Observable implications** (of intrinsic human stochasticity)

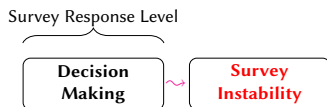
Sources of Stochasticity Leading to Survey Instability



Used to find

- **Observable implications** (of intrinsic human stochasticity)
- **Practical implications**

Sources of Stochasticity Leading to Survey Instability



Used to find

- **Observable implications** (of intrinsic human stochasticity)
- **Practical implications**

Stochasticity Sources: Decision Making

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed).

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices 

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices 

Election Day

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices 

Election Day

- 10 offices, each with Party A & B candidates

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices 

Election Day

- 10 offices, each with Party A & B candidates
- In 6/10, Party A is better for you than Party B.

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices 

Election Day

- 10 offices, each with Party A & B candidates
- In 6/10, Party A is better for you than Party B.
- How many Party A candidates will you choose?

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices 

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

C_t

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \left\{ \right.$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \left\{ \begin{array}{l} = \mathbf{1}(\pi > 0.5) \end{array} \right.$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \left\{ \begin{array}{l} = \mathbf{1}(\pi > 0.5) \end{array} \right. \quad \text{Expected Utility (deterministic)}$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \left\{ \begin{array}{l} = \mathbf{1}(\pi > 0.5) \quad \text{Expected Utility (deterministic)} \quad E(D) = 0 \end{array} \right.$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) \\ = \mathbf{1}(\pi + \eta_t > 0.5) \end{cases} \quad \text{Expected Utility (deterministic)} \quad E(D) = 0$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) \end{cases}$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \end{cases}$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) \end{cases}$$

Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & \end{cases}$$

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- **Literature:** Probability matching! (humans, bees, fish, pigeons...)

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- Literature: Probability matching! (humans, bees, fish, pigeons...)
- Experiments:

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- Literature: Probability matching! (humans, bees, fish, pigeons...)
- Experiments:
 - Tell (teach) respondents their beliefs (π)

Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- **Literature:** Probability matching! (humans, bees, fish, pigeons...)
- **Experiments:**
 - **Tell (teach) respondents their beliefs (π)**
 - **Minimize:** η , uncertainty, indifference

Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- **Literature:** Probability matching! (humans, bees, fish, pigeons...)
- **Experiments:**
 - **Tell (teach) respondents their beliefs (π)**
 - **Minimize:** η , uncertainty, indifference
 - **Result:** **still** 40-60% stochastic!

Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- **Literature:** Probability matching! (humans, bees, fish, pigeons...)
- **Experiments:**
 - **Tell (teach) respondents their beliefs (π)**
 - **Minimize:** η , uncertainty, indifference
 - **Result:** **still** 40-60% stochastic! 🙅

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)

- Beliefs: $\theta \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$ (unknown)

- Cost

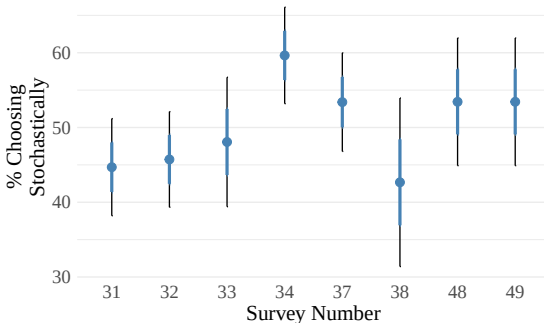
- History

C_t {

- Learning

- Exploration

Decision Making: Highly Stochastic!



Stochasticity Sources: Decision Making

- **Preference:** $\rho \in \{0, 1\}$ (unknown)
- **Belief:** $\pi = \Pr(\rho = 1)$ (known)
- **Choices:** C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- **How Beliefs become Choices**

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- **Literature:** Probability matching! (humans, bees, fish, pigeons...)
- **Experiments:**
 - **Tell (teach) respondents their beliefs (π)**
 - **Minimize:** η , uncertainty, indifference
 - **Result:** **still** 40-60% stochastic! 🙅

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)
- Belief: $\pi = \Pr(\rho = 1)$ (known)
- Choices: C_1, C_2 (observed). [Instability: $D = \mathbf{1}(C_1 \neq C_2)$]
- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

- Literature: Probability matching! (humans, bees, fish, pigeons...)
- Experiments:
 - Tell (teach) respondents their beliefs (π)
 - Minimize: η , uncertainty, indifference
 - Result: still 40-60% stochastic!

$\Rightarrow$ Human Decision Making: Adds Stochasticity

Stochasticity Sources: Cognitive Processing

Stochasticity Sources: Cognitive Processing

**Survey
Instability**

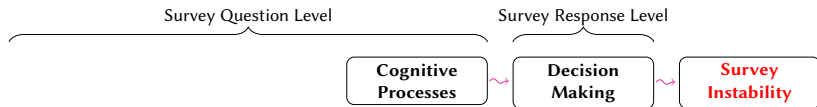
Stochasticity Sources: Cognitive Processing

Survey Response Level

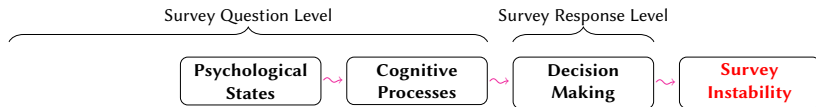
**Decision
Making**

**Survey
Instability**

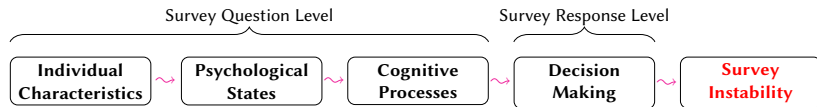
Stochasticity Sources: Cognitive Processing



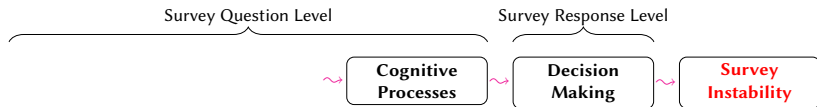
Stochasticity Sources: Cognitive Processing



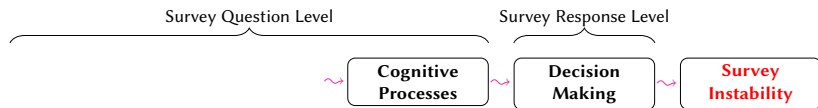
Stochasticity Sources: Cognitive Processing



Stochasticity Sources: Cognitive Processing

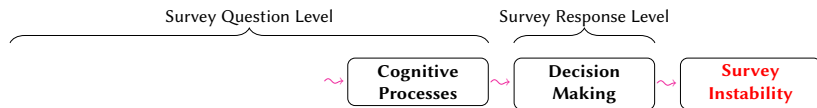


Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

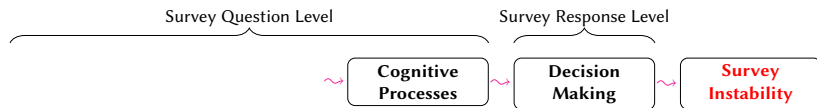
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- **High cognitive complexity:** hard Qs; many As

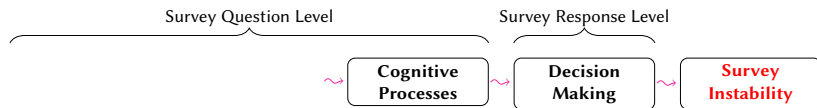
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- **High cognitive complexity:** hard Qs; many As
- **Low time-on-task:** for any given question

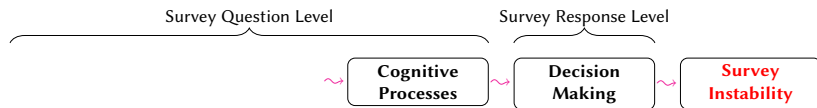
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- **High cognitive complexity:** hard Qs; many As
- **Low time-on-task:** for any given question
- **Differential cognitive processing:** Q_1 vs Q_2

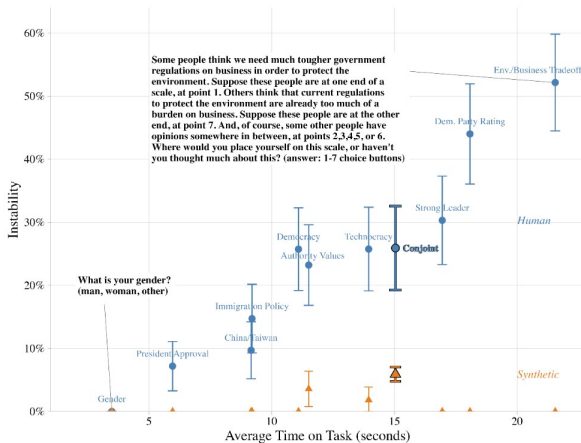
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- High cognitive complexity: hard Qs; many As 📌
- Low time-on-task: for any given question
- Differential cognitive processing: Q_1 vs Q_2

Stochasticity Sources: Cognitive Processing

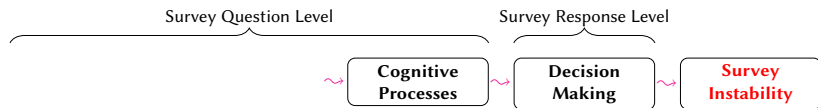


Key
ility

What

- H
- L
- D

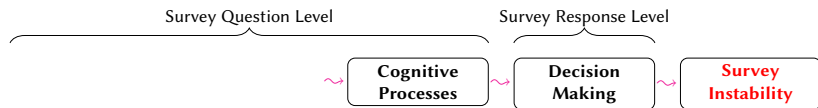
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- High cognitive complexity: hard Qs; many As 📖
- Low time-on-task: for any given question
- Differential cognitive processing: Q_1 vs Q_2

Stochasticity Sources: Cognitive Processing

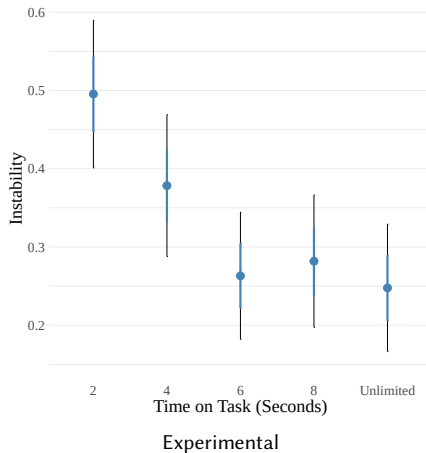
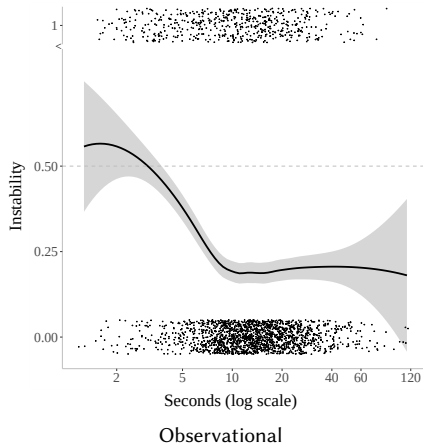


What increases stochasticity (and thus instability)?

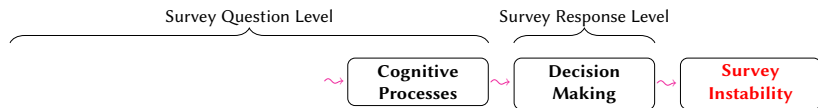
- High cognitive complexity: hard Qs; many As
- Low time-on-task: for any given question 📱
- Differential cognitive processing: Q_1 vs Q_2

Stochasticity Sources: Cognitive Processing

Limited time-on-task causes survey instability



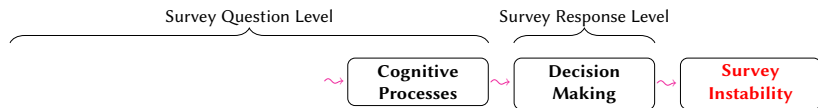
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- High cognitive complexity: hard Qs; many As
- Low time-on-task: for any given question 📱
- Differential cognitive processing: Q_1 vs Q_2

Stochasticity Sources: Cognitive Processing

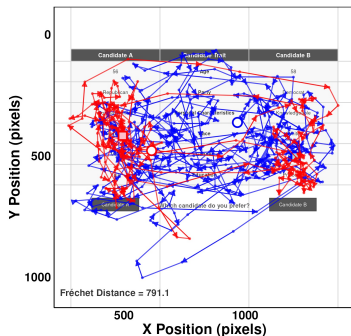


What increases stochasticity (and thus instability)?

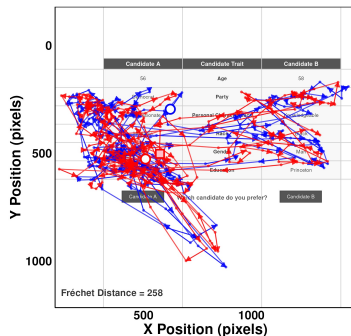
- **High cognitive complexity:** hard Qs; many As
- **Low time-on-task:** for any given question
- **Differential cognitive processing:** Q_1 vs Q_2 🖐️

Stochasticity Sources: Cognitive Processing

Stochasticity in Gaze Behavior

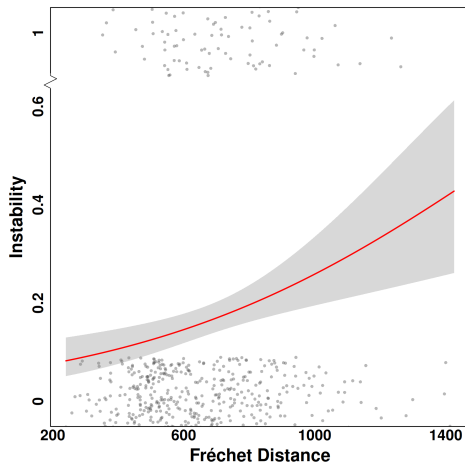


Large difference



Small difference

Stochastic Distance $\leadsto$ Survey Instability

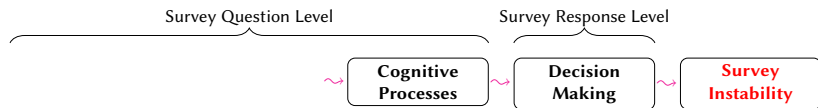


What

- H
- L
- D

vey
bility

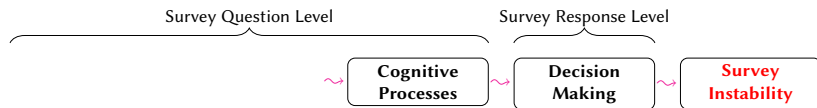
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- **High cognitive complexity:** hard Qs; many As
- **Low time-on-task:** for any given question
- **Differential cognitive processing:** Q_1 vs Q_2 🖐️

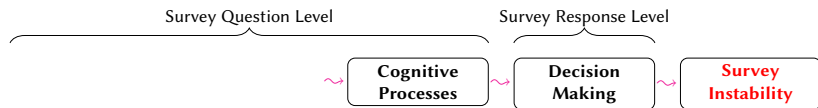
Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- **High cognitive complexity:** hard Qs; many As
- **Low time-on-task:** for any given question
- **Differential cognitive processing:** Q_1 vs Q_2

Stochasticity Sources: Cognitive Processing



What increases stochasticity (and thus instability)?

- High cognitive complexity: hard Qs; many As
- Low time-on-task: for any given question
- Differential cognitive processing: Q_1 vs Q_2

⇒ Human Cognitive Processing: Adds Stochasticity

Stochasticity Sources: Psychological States

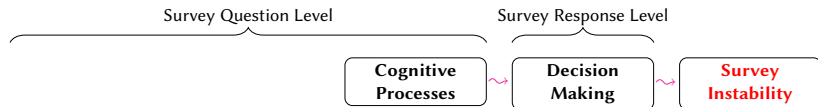
**Survey
Instability**

Stochasticity Sources: Psychological States

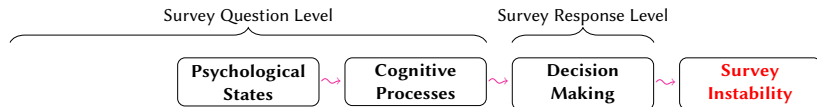
Survey Response Level



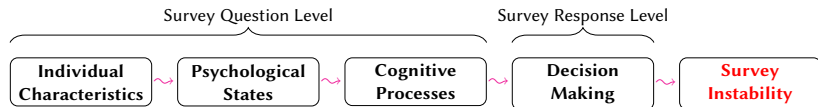
Stochasticity Sources: Psychological States



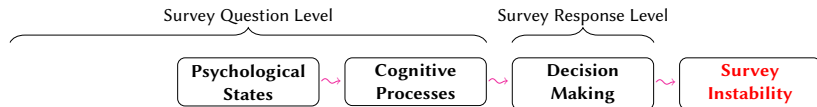
Stochasticity Sources: Psychological States



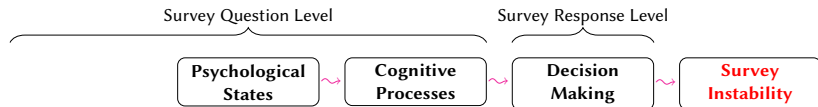
Stochasticity Sources: Psychological States



Stochasticity Sources: Psychological States

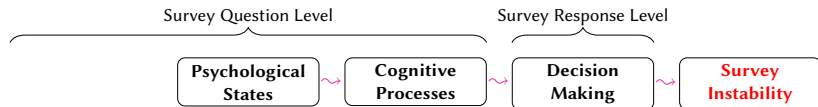


Stochasticity Sources: Psychological States



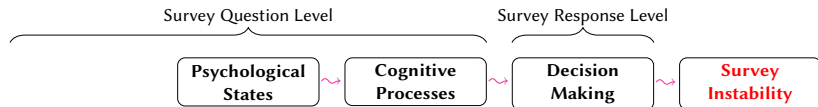
- Psychological States Summary

Stochasticity Sources: Psychological States



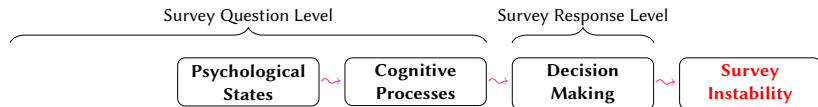
- **Psychological States Summary**
 - **Attention Checks** ruin surveys

Stochasticity Sources: Psychological States



- **Psychological States Summary**
 - **Attention Checks** ruin surveys
 - Use **Preoccupation** instead

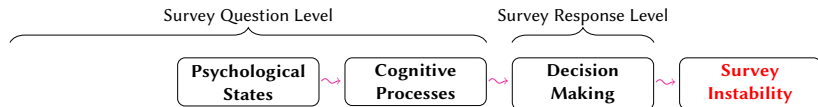
Stochasticity Sources: Psychological States



- **Psychological States Summary**
 - **Attention Checks** ruin surveys
 - Use **Preoccupation** instead

Everyone comes to a new task (like this survey) with things on their minds. How preoccupied were you, just before you started this survey, with thoughts or worries about your partner, job, health, children, friends, money, or other concerns?

Stochasticity Sources: Psychological States

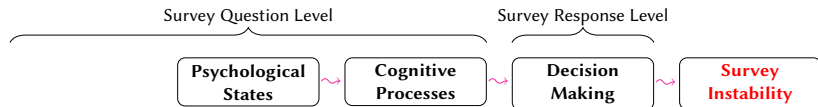


- **Psychological States Summary**
 - **Attention Checks** ruin surveys
 - Use **Preoccupation** instead

Everyone comes to a new task (like this survey) with things on their minds. How preoccupied were you, just before you started this survey, with thoughts or worries about your partner, job, health, children, friends, money, or other concerns?

{Very, Fairly, Slightly, Not at all} Preoccupied

Stochasticity Sources: Psychological States



- **Psychological States Summary**
 - **Attention Checks** ruin surveys
 - Use **Preoccupation** instead

Everyone comes to a new task (like this survey) with things on their minds. How preoccupied were you, just before you started this survey, with thoughts or worries about your partner, job, health, children, friends, money, or other concerns?

{Very, Fairly, Slightly, Not at all} Preoccupied

⇒ **Human Psychology: Adds Stochasticity**

So Who's to Blame for Survey Instability?

So Who's to Blame for Survey Instability?

Explanations

So Who's to Blame for Survey Instability?

Explanations

Survey Instability

~25%

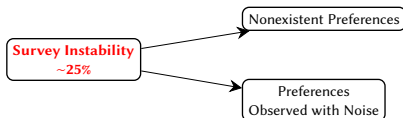
So Who's to Blame for Survey Instability?

Explanations



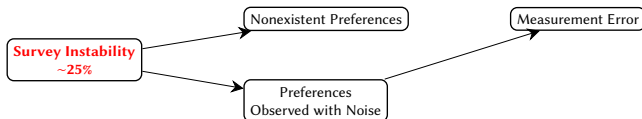
So Who's to Blame for Survey Instability?

Explanations



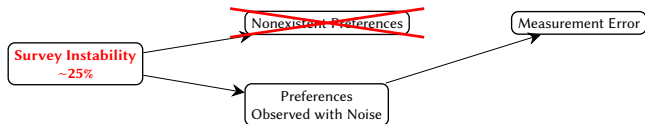
So Who's to Blame for Survey Instability?

Explanations



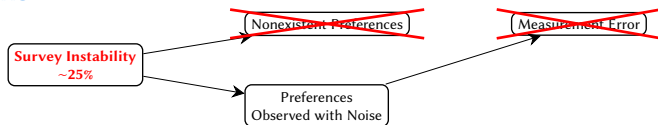
So Who's to Blame for Survey Instability?

Explanations



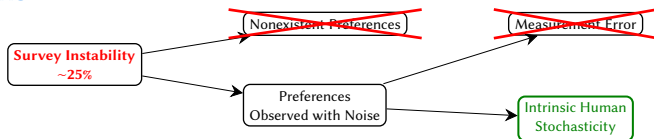
So Who's to Blame for Survey Instability?

Explanations



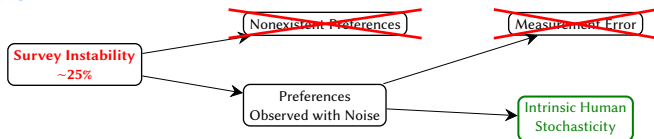
So Who's to Blame for Survey Instability?

Explanations



So Who's to Blame for Survey Instability?

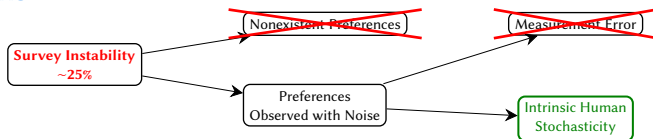
Explanations



Observable Implications

So Who's to Blame for Survey Instability?

Explanations

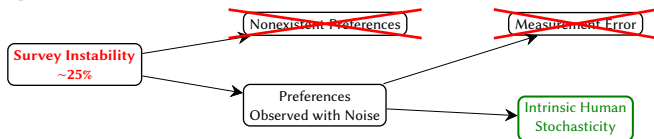


Observable Implications

Survey
Instability

So Who's to Blame for Survey Instability?

Explanations

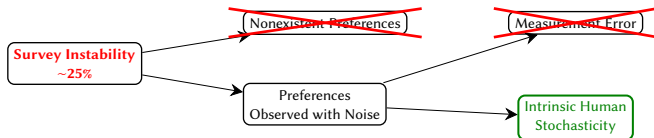


Observable Implications

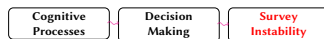


So Who's to Blame for Survey Instability?

Explanations

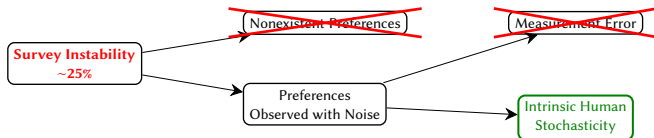


Observable Implications



So Who's to Blame for Survey Instability?

Explanations

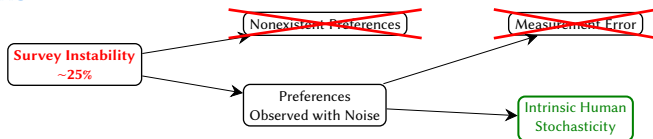


Observable Implications

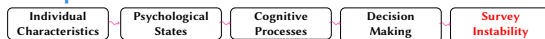


So Who's to Blame for Survey Instability?

Explanations

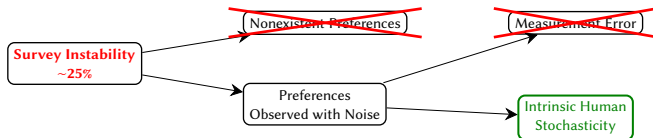


Observable Implications

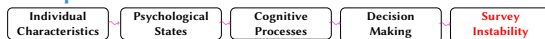


So Who's to Blame for Survey Instability?

Explanations



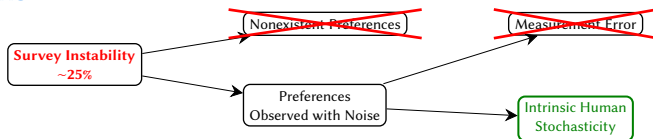
Observable Implications



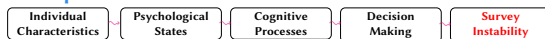
Practical Implications

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

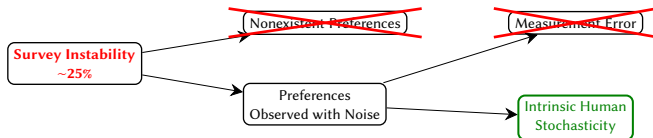


Practical Implications

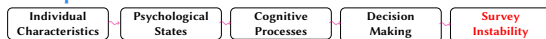
- Improve survey instruments:

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

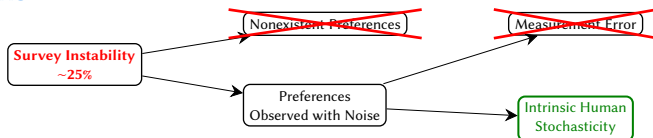


Practical Implications

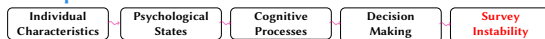
- Improve survey instruments:
 - measure instability (one extra Q)

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

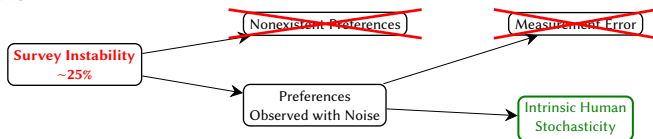


Practical Implications

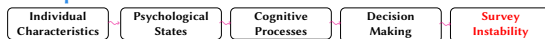
- Improve survey instruments:
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

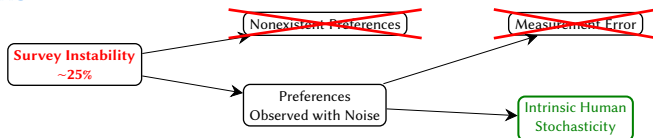


Practical Implications

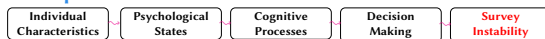
- Improve survey instruments:
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- Filter likely-liable respondents:

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

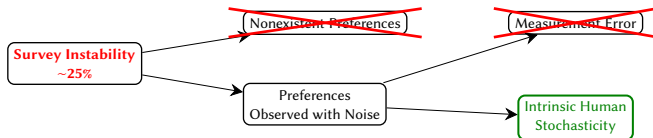


Practical Implications

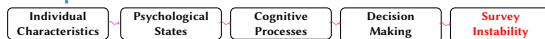
- **Improve survey instruments:**
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- **Filter likely-unsafe respondents:**
 - preoccupied (no attention checks)

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

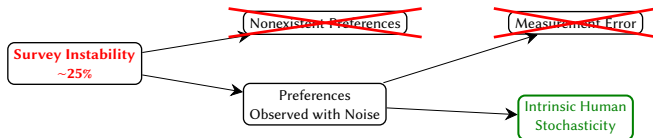


Practical Implications

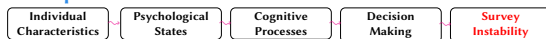
- **Improve survey instruments:**
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- **Filter likely-unsafe respondents:**
 - preoccupied (no attention checks)
 - low time-on-task

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

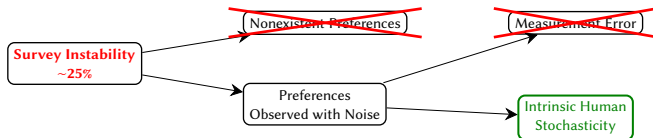


Practical Implications

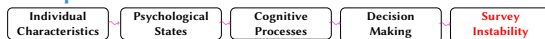
- Improve survey instruments:
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- Filter likely-unsafe respondents:
 - preoccupied (no attention checks)
 - low time-on-task
- Improve statistical methods:

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

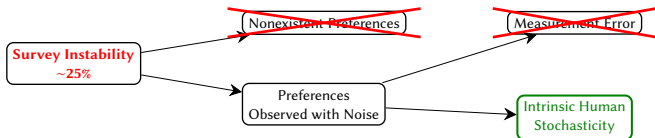


Practical Implications

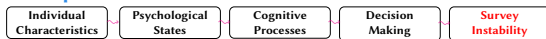
- **Improve survey instruments:**
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- **Filter likely-unsafe respondents:**
 - preoccupied (no attention checks)
 - low time-on-task
- **Improve statistical methods:**
 - selection (change QOI, oversample filtered groups, or impute)

So Who's to Blame for Survey Instability?

Explanations



Observable Implications

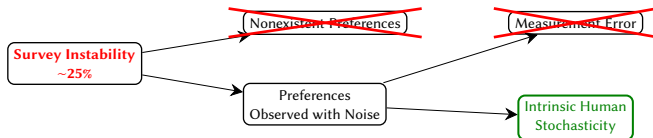


Practical Implications

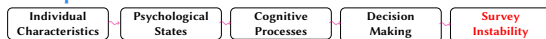
- **Improve survey instruments:**
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- **Filter likely-unsafe respondents:**
 - preoccupied (no attention checks)
 - low time-on-task
- **Improve statistical methods:**
 - selection (change QOI, oversample filtered groups, or impute)
 - remaining instability (correct statistically)

So Who's to Blame for Survey Instability?

Explanations



Observable Implications



Practical Implications

- **Improve survey instruments:**
 - measure instability (one extra Q)
 - reduce cognitive complexity (simpler Qs, fewer As)
- **Filter likely-unsafe respondents:**
 - preoccupied (no attention checks)
 - low time-on-task
- **Improve statistical methods:**
 - selection (change QOI, oversample filtered groups, or impute)
 - remaining instability (correct statistically)